

Here is the technology-agnostic Technical Design Specification for V1, derived strictly from the canonical domain model, blueprints, and the ruthless scope cut.

1. System Modules & Aggregate Ownership

The V1 system is scoped strictly to a single issuer operating manually through a UI, handling only FE (01) and CCF (03) documents, with a focus on immutable evidence, auditability, and surviving connectivity drops.

- **Document Type Registry**
 - *Internal Responsibilities:* Acts as the fixed configuration seam for document rules, pinning required fields, tax behavior, schema versions, and receptor rules.
 - *Owned Aggregates:* Document Type Specification (01 and 03 only in V1).
- **Issuance**
 - *Internal Responsibilities:* Manages the composition, validation, identification, signing, and transmission of fiscal documents. Handles the core state machine.
 - *Owned Aggregates:* FiscalDocument, ControlNumberSeries, TransmissionRecord.
- **Invalidation**
 - *Internal Responsibilities:* Lawfully voids an already-accepted document via plain invalidation (without replacement).
 - *Owned Aggregates:* InvalidationEvent.
- **Contingency**
 - *Internal Responsibilities:* Governs offline issuance periods, flagging documents as held, and orchestrating batch reconciliation once connectivity returns.
 - *Owned Aggregates:* ContingencyEpisode.
- **Parties & Counterparties**
 - *Internal Responsibilities:* Maintains reusable receptor master data and validates tax-identity well-formedness.
 - *Owned Aggregates:* Counterparty.
- **Product & Catalog**
 - *Internal Responsibilities:* Maintains reusable item master data, default unit prices, and tax behaviors.
 - *Owned Aggregates:* Item.
- **Evidence & Archive**
 - *Internal Responsibilities:* Preserves an immutable, append-only proof of everything issued, including payloads, stamps, and representations.
 - *Owned Aggregates:* EvidenceBundle.
- **Audit Trail**
 - *Internal Responsibilities:* Records a tamper-evident, hash-chained history of all significant system and business actions.
 - *Owned Aggregates:* AuditTrail.
- **Access & Identity**
 - *Internal Responsibilities:* Authenticates human actors and authorizes actions based on exactly two roles (Owner, Biller).
 - *Owned Aggregates:* User.

2. Business Operations

2.1 Commands (State-Mutating Actions)

- **Issuance Commands:** CreateDraft, UpdateDraft, ValidateDocument, SignDocument, TransmitDocument.
- **Invalidation Commands:** RequestInvalidation, TransmitInvalidation.
- **Contingency Commands:** OpenEpisode, EnrollDocument, TransmitNotification, CloseEpisode.
- **Master Data Commands:** CreateCounterparty, UpdateCounterparty, CreateItem, UpdateItem.
- **System Commands:** AssignUserRole, ConfigureIssuer.

2.2 Queries (Read Operations)

- **Document Queries:** GetDocumentByIdentity, ListDocuments (with filters).
- **Operational Queries:** GetOpenContingencyEpisode, CheckControlNumberAvailability.
- **Evidence/Audit Queries:** ViewEvidenceBundle, DownloadHumanReadableRepresentation, ViewAuditTrail.
- **Master Data Queries:** ListCounterparties, ListItems.

2.3 Domain Services

- **Identity Assignment Service:** Atomically assigns a globally unique Generation Code and a gapless Control Number to a document upon validation.
- **Cryptographic Orchestration Service:** Freezes the canonical payload and captures the immutable Issuer/Receptor snapshots to legally form the document.
- **Representation Generation Service:** Deterministically generates the Human-Readable Representation (PDF) from the signed payload.

3. Validation Rules (Invariants)

- **Type-Driven Rule Evaluation:** A document's DTE Type (01 or 03) is fixed at creation. CCFs require a fully identified taxpayer Receptor Snapshot; FEs allow relaxed identities.
- **Reconciliation Integrity:** Document Totals must perfectly equal the sum of Document Lines plus the Tax Summary before a document can leave the Draft state.
- **Monotonic Numbering:** Control Number Series are gapless and strictly monotonic; consumed correlatives are never re-issued, even if a document is rejected or invalidated.
- **Immutable Snapshots:** Once Signed, the Signed Payload, Receptor Snapshot, and Issuer Snapshot become permanently immutable; they do not update if the underlying Counterparty or Item master data changes.
- **Irreversible Signing:** Any correction to a Signed document requires creating an entirely new Fiscal Document.

4. Cross-Module Interactions

- **Issuance $\rightarrow$ Type Registry:** Issuance reads the pinned Schema Version,

required fields, and rules from the Registry during the validation phase.

- **Issuance $\rightarrow$ Catalog/Parties:** Issuance requests data from Counterparties and Items to copy them into the immutable Receptor and Document Line snapshots.
- **Issuance $\rightarrow$ Evidence & Archive:** Irreversible domain facts (signing, acceptance) in Issuance trigger Evidence & Archive to append artifacts (payloads, stamps, hashes) to the Evidence Bundle.
- **Invalidation $\rightarrow$ Issuance:** Upon an Invalidation Event being accepted by the authority, the Invalidation module commands Issuance to transition the target Fiscal Document to the Invalidated state.
- **All State-Changing Modules $\rightarrow$ Audit Trail:** Every state transition, transmission outcome, access-control decision, and master data edit must append a hash-chained entry to the Audit Trail.

5. Application Workflows & User Journeys

- **Manual Issuance (Happy Path):**
 1. Actor creates a Draft and edits lines/receptors.
 2. Actor requests Validation; system assigns Generation Code and Control Number.
 3. Actor requests Signing; payload freezes, Evidence Bundle is created.
 4. System transmits to Authority.
 5. Authority returns an accepted outcome; system stores the Acceptance Stamp, Document is legally valid.
- **Contingency (Offline Issuance):**
 1. Authority unreachable; Contingency Episode is Opened.
 2. Actor drafts, validates, and signs documents normally; documents are placed in Contingency-Held status.
 3. Connectivity restores; episode transitions to Reconciling.
 4. System transmits contingency notification first, followed by held documents within the legal window.
 5. Episode Closes when all reach a terminal outcome.
- **Plain Invalidation:**
 1. Actor initiates invalidation on an Accepted document.
 2. System creates an Invalidation Event (reason type: rescind/other).
 3. Event is Submitted.
 4. If Accepted by authority, the target document status changes to Invalidated; invalidation evidence is stored.

6. Screen Inventory

Derived strictly from the V1 scope of UI-driven, single-issuer manual operations.

1. **Composer (Issuance Form):** The primary UI for creating and editing a Draft. Type-aware to enforce FE vs. CCF rules.
2. **Document Ledger:** List view with filters for exploring drafted, valid, signed, accepted, rejected, and invalidated documents.
3. **Document Details Viewer:** Displays the aggregate state, evidence artifacts, validation

findings, and provides the "Download PDF" and "Download JSON" capabilities.

4. **Catalog Manager (Items & Counterparties):** Basic, flat CRUD interfaces for creating and maintaining reusable product and receptor records.
5. **Contingency Dashboard:** Interface to manually declare an episode, view held documents, and trigger the manual batch resend upon connectivity restoration.
6. **Invalidation Modal/Flow:** UI to select an accepted document, provide a justification, and submit the plain invalidation request.

7. Error Scenarios

- **Authority Rejection (Terminal):** If a document is Submitted and the authority refuses it, the status becomes Rejected. The document is dead; the Control Number is considered consumed. The user must create a brand-new document to correct it.
- **Invalidation Rejection:** If an invalidation request is refused, the target document remains legally Accepted. The user may attempt a new invalidation.
- **Validation Totals Mismatch:** If Document Totals do not equal the sum of lines plus tax summary, the document is blocked from leaving the Draft state.
- **Contingency Window Breach:** If held documents exceed the legal window before reconciliation, the rule breach is recorded and escalated.

8. Permission Matrix

V1 contains exactly two human roles.

Capability	Owner	Billor
Manage users, roles, and configuration	Allowed	Denied
Manage catalog and counterparties	Allowed	Denied
Compose, sign, and transmit documents	Allowed	Allowed
Invalidate documents	Allowed	Allowed

9. Reporting Requirements

- **Tax-Book Export (Libro de Ventas):** A simple flat-file CSV export of issued documents to satisfy monthly tax filing needs.
- **Audit Extraction:** While V1 does not feature a complex audit-browsing UI, the system must support exposing the append-only, hash-chained Audit Trail (likely via SQL/export)

for regulatory inspection.